

# tests/letsencrypt/fixtures/ gen\_fixtures.sh — operator guide

**Revision:** 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**  
Active hermetic openssl generator (§11.4.77 regeneration  
mechanism) for the cert\_analyzer golden-good / golden-bad fixture  
corpus.

Companion (§11.4.18) to the in-source documentation block  
at the top of the script.

## Overview / Purpose

Regenerates the golden-good + golden-bad certificate fixtures  
that self-validate cert\_analyzer.sh (§11.4.107(10)). Everything it  
produces is EPHEMERAL throwaway test material — self-signed  
TEST CAs, NOT real Let's Encrypt certs, NO production private  
keys, NO secrets (§11.4.10). The generated \*.key / \*.csr / \*.srl are  
gitignored (tests/letsencrypt/fixtures/.gitignore); only the public  
\*.pem certs are tracked and regenerated on demand by re-running  
this script (§11.4.77).

Fully deterministic: FIXED validity windows are baked in via  
openssl -not\_before / -not\_after so the fixtures do NOT depend on  
the generation date — they are pinned relative to the reference  
"now" 2026-07-01T12:00:00Z used by the self-test (§11.4.50).

## Usage

```
GOMAXPROCS=2 nice -n 19 ionice -c 3 \  
bash tests/letsencrypt/fixtures/gen_fixtures.sh
```

Runs from anywhere — it resolves its own directory and cds into  
it. Normally invoked automatically by cert\_analyzer\_selftest.sh  
when the tracked \*.pem set is missing.

## Inputs

None (fully deterministic — validity windows are hard-coded constants).

## Outputs

Under tests/letsencrypt/fixtures/ (tracked \*.pem):

| Fixture             | Content                                                                               |
|---------------------|---------------------------------------------------------------------------------------|
| test_ca.pem         | self-signed TEST CA (the "expected CA").                                              |
| good_leaf.pem       | issued by test_ca, SAN proxy.test, window straddling the ref now (~60d remain).       |
| expired_leaf.pem    | issued by test_ca, NotAfter in the past.                                              |
| nearexpiry_leaf.pem | issued by test_ca, ~4-5 days left at ref now.                                         |
| wrongca_leaf.pem    | signed by a DIFFERENT CA, SAN proxy.test (right host, WRONG issuer).                  |
| otherhost_leaf.pem  | issued by test_ca, SAN other.test (right issuer, WRONG host).                         |
| wildcard_leaf.pem   | issued by test_ca, SAN *.proxy.test.                                                  |
| doublewild_leaf.pem | issued by test_ca, SAN *.*.proxy.test (malformed double-wildcard — must never match). |
| nosan_leaf.pem      | issued by test_ca, CN proxy.test, NO SAN extension (empty-SAN).                       |
| ipsan_leaf.pem      | issued by test_ca, SAN IP:10.0.0.1 only (IP-only SAN).                                |
| dnsandip_leaf.pem   | issued by test_ca, SAN DNS:proxy.test, IP:10.0.0.1 (mixed).                           |
| malformed_leaf.pem  | present-but-truncated/unparseable PEM (first 3 lines of good_leaf.pem).               |

Plus gitignored \*.key / \*.csr / \*.srl throwaway material. Emits a gen\_fixtures: wrote <list> line on stdout.

## Side-effects

Writes the files above into tests/letsencrypt/fixtures/; removes the throwaway serial files (test\_ca.srl, wrong\_ca.srl) and each intermediate \*.csr. No network, no containers. set -eu (aborts on error / unset var).

## Dependencies

POSIX sh, openssl — specifically req -x509 / x509 -req with -not\_before / -not\_after / -copy\_extensions / -no\_check\_time (**OpenSSL 3.2+**), plus head, rm, printf. POSIX-clean — parses under sh -n AND bash -n (§11.4.67).

## Edge cases

- **Malformed fixture** is produced deterministically by truncating good\_leaf.pem to its first 3 lines — guaranteed unparseable regardless of key bytes.
- **Wildcard fixtures** exercise the single-leading-wildcard branch (\*.proxy.test) and the illegal double-wildcard reject (\*.\*.proxy.test).
- **Empty-SAN / IP-only / mixed DNS+IP** fixtures pin the SAN-matching discipline (no CN fallback; dNSName-only; DNS isolated from IP).
- Re-running is idempotent — it overwrites the tracked \*.pem set with byte-equivalent windows (dates are fixed; only the throwaway keys differ).

## Related scripts

- tests/letsencrypt/cert\_analyzer.sh — the library these fixtures validate.
- tests/letsencrypt/cert\_analyzer\_selftest.sh — consumes the fixtures and self-bootstraps by running this generator when they are absent.
- Constitution §11.4.77 (regeneration mechanism), §11.4.10 (no real secrets), §11.4.107(10) (self-validated analyzer), §11.4.50; design LETSENCRYPT\_HTTPS\_PLAN.md Phase 3.

## Last verified

2026-07-01 — documented against the script source; sh -n / bash -n parse-clean. Produces the tracked \*.pem corpus consumed by cert\_analyzer\_selftest.sh (requires OpenSSL 3.2+ for the date flags).